



# Principles of Finance

# Chapter 15

## End-of-Chapter Questions

How to Think About Investing

### Multiple Choice Questions

1. The total dollar return equals \_\_\_\_\_.
- A. the EPS of a stock
  - B. capital gains income plus dividend income
  - C. the price paid for a share of stock minus the selling price of the stock
  - D. the price paid for a share of stock divided by the selling price of the stock

Solution: B

#### Section 15.1 LO 1 Easy

2. The dividend yield is calculated by \_\_\_\_\_.
- A. dividing the price of the stock by the EPS
  - B. subtracting any capital loss from the capital gain
  - C. dividing the annual dividend by the initial stock price
  - D. dividing the annual dividend by the net income for the year

Solution: C

#### Section 15.1 LO 1 Easy

3. Which of the following is the best example of firm-specific risk?
- A. A global pandemic causes major disruptions in the economy.
  - B. The Federal Reserve increases the money supply dramatically, leading to massive inflation.
  - C. AAA Pharmaceuticals withdraws a medication as it studies whether strokes five people suffered after taking the medication were related to the medication.
  - D. As an arctic blast descends on North America, most of the United States is blanketed in snow or ice.

Solution: C

#### Section 15.1 LO 3 Medium

4. Investors diversify their portfolio in order to \_\_\_\_\_.
- A. reduce risk
  - B. increase risk
  - C. increase return
  - D. increase the standard deviation

Solution: A

#### Section 15.2 LO 1 Easy

5. Which of the following would be the best example of systematic risk?
- A. An error in the company's computer system miscalculates the amount of inventory that Monique's Boutique is holding.
  - B. BlueJay Air has a reduction in new reservations following a crash of one of its jets.
  - C. The spokesperson for Serena's Sports Shoes is involved in an ethical scandal.
  - D. Interest rates rise after the Federal Reserve announces it will slow down the rate of growth of the money supply.

Solution: D

**Section 15.2 LO 3 Medium**

6. As the number of stocks in a portfolio increases, \_\_\_\_\_.
- A. firm-specific risk increases
  - B. systematic risk becomes zero
  - C. systematic risk decreases and returns increase
  - D. firm-specific risk is reduced but systematic risk remains

Solution: D

**Section 15.2 LO 4 Medium**

7. Beta is a measure of \_\_\_\_\_.
- A. systematic risk
  - B. firm-specific risk
  - C. a firm's profitability
  - D. a stock's dividend yield

Solution: A

**Section 15.3 LO 2 Easy**

8. Which of the following would be the best estimate of the risk-free rate?
- A. The rate of inflation
  - B. The average return on the S&P 500
  - C. The average return on Amazon's stock
  - D. The average return on US Treasury bills

Solution: D

**Section 15.3 LO 1 Easy**

9. The Sharpe ratio can be considered a measure of \_\_\_\_\_.
- A. the reward of an investment in relation to the risk
  - B. the systematic risk of a stock
  - C. the total return of a stock investment
  - D. the historical return of an individual security

Solution: A

**Section 15.4 LO 1 Easy**

10. A positive Jensen's alpha indicates that a portfolio has \_\_\_\_\_.

- A. a negative beta
- B. an abnormal return
- C. more total risk than the average portfolio
- D. more systematic risk than the average portfolio

Solution: B

**Section 15.4 LO 3 Medium**

11. If an equally weighted portfolio contains 10 stocks, then \_\_\_\_\_.

- A. the stocks in the portfolio will each have a weight of 0.10
- B. the return of the portfolio must be multiplied by 10 to get the annualized return
- C. the standard deviation of the portfolio will be one-tenth the standard deviation of one of the stocks
- D. the standard deviation of the portfolio will be 10 times the standard deviation of one of the stocks

Solution: A

**Section 15.5 LO 2 Difficult**

## Review Questions

1. What is the difference between firm-specific risk and unsystematic risk?

Solution: Firm-specific risks impact a particular company. These risks would impact the expected revenues or expenses of the company. Examples include a company with a product liability lawsuit brought against it or a major explosion destroying a company's manufacturing facility. Unsystematic risk is known as market risk; it impacts all companies and industries.

**Section 15.2 LO 3 Easy**

2. Explain why diversification reduces unsystematic risk but not systematic risk.

Solution: Diversification means that a smaller percentage of an investor's money is impacted when a firm-specific event occurs; the impact of that event is diluted as an investor diversifies across many companies. Investors are still exposed to the overall market conditions and economy, no matter how diversified they are.

**Section 15.2 LO 1 Easy**

3. Explain what happens to the standard deviation of returns of a portfolio as the number of stocks in the portfolio increases.

Solution: As the number of stocks in a portfolio increases, the standard deviation of the portfolio declines. This is because the movement of one stock is not perfectly correlated with the movement of other stocks. As the number of stocks in a portfolio continues to rise, the size of the drop in the standard deviation from adding one more stock decreases. The standard deviation never goes to zero because the investor still faces systematic risk.

**Section 15.2 LO 4 Medium**

4. Enrique owns five stocks: Alaska Airlines, American Airlines, Delta Airlines, Southwest Airlines, and Ford. Radha also owns five stocks: Apple, McDonald's, Tesla, Facebook, and Disney. Does Enrique or Radha have a more diversified portfolio?

Solution: Radha has a more diversified portfolio than Enrique does. Although they both own five stocks, four of Enrique's stocks are airlines. While each of these airlines has its own risks (such as a negative passenger experience with an employee being posted on social media), they share common risks such as oil prices and weather conditions. Radha owns stocks in different industries, so the stocks she owns will be less correlated.

**Section 15.2 LO 1 Medium**

5. You are considering purchasing shares in a company that has a beta of 0.8. Explain what this beta means.

Solution: A beta less than 1 means that the stock is less risky than the average stock in the market. For every 1% move in the market, this company tends to move by only 0.8%.

**Section 15.3 LO 2 Medium**

6. Explain how the Sharpe ratio and the Treynor ratio can be considered reward-to-risk measures.

Solution: Both the Sharpe ratio and the Treynor ratio measure how much the return of a portfolio exceeds the return on a risk-free security. The denominator of the Sharpe ratio is the standard deviation of the portfolio, which is a measure of the portfolio's volatility, or how risky it is. The denominator of the Treynor ratio is the beta of the portfolio, which is a measure of how risky the portfolio is compared to the market portfolio. Therefore, both ratios measure the reward relative to the amount of risk the portfolio investors take on.

**Section 15.4 LO 1 Medium**

## Problems

1. You purchase 100 shares of COST (Costco) for \$280 per share. Three months later, you sell the stock for \$290 per share. You receive a dividend of \$0.57 a share. What is your total dollar return?

Solution:

$$\text{Total Dollar Return} = \$290.00 + \$0.57 - \$280 = \$10.57$$

**Section 15.1 LO 1 Easy**

2. You purchase 100 shares of COST for \$280 per share. Three months later, you sell the stock for \$290 per share. You receive a dividend of \$0.57 a share. What are your dividend yield, capital gain yield, and total percentage return?

Solution:

$$\text{Dividend Yield} = \frac{\text{Dividend}}{\text{Stock Purchase Price}} = \frac{\$0.57}{\$280} = 0.20\%$$

$$\text{Capital Gain Yield} = \frac{\text{Stock Sale Price} - \text{Stock Purchase Price}}{\text{Stock Purchase Price}} = \frac{\$290 - \$280}{\$280} = 3.57\%$$

$$\text{Total Percentage Return} = \text{Dividend Yield} + \text{Capital Gain Yield} = 0.20\% + 3.57\% = 3.77\%$$

#### Section 15.1 LO 1 Medium

3. You purchase 100 shares of COST for \$280 per share. Three months later, you sell the stock for \$290 per share. You receive a dividend of \$0.57 a share. What is the EAR of your investment?

Solution:

$$\begin{aligned} \text{EAR} &= (1 + \text{Holding Period Percentage Return})^m - 1 \\ &= (1 + 0.03775)^4 - 1 \\ &= 1.15975 - 1 = 0.15976 \text{ or } 15.976\% \end{aligned}$$

#### Section 15.1 LO 1 Medium

4. You invest in a stock for four years. The returns for the four years are 20%, -10%, 15%, and -5%. Calculate the arithmetic average return and the geometric average return.

Solution:

$$\text{Arithmetic Average} = \frac{0.2 + (-0.1) + 0.15 + (-0.05)}{4} = 5\%$$

$$\text{Geometric Average} = [(1 + 0.2)(1 - 0.1)(1 + 0.15)(1 - 0.05)]^{0.25} - 1 = 4.22\%$$

#### Section 15.1 LO 2 Medium

5. You are considering purchasing shares in a company that has a beta of 0.9. The average return for the S&P 500 is 11%, and the average return for US Treasury bills has been 2%. Based on the CAPM, what is your expected return for the stock?

Solution:

$$\begin{aligned} R_e &= R_f + \text{Beta} \times \text{Market Risk Premium} \\ R_e &= R_f + \text{Beta} \times (R_m - R_f) = R_{\text{US T-bill}} + \text{Beta} \times (R_{\text{S\&P 500}} - R_{\text{US T-bill}}) \\ R_e &= 0.02 + 0.9(0.11 - 0.02) = 0.02 + 0.081 = 0.101 = 10.1\% \end{aligned}$$

The expected return for the stock is 10.1%, slightly less than the average market return of 11%, because the stock has a beta of less than 1 and is slightly less risky than the average stock in the market.

#### Section 15.3 LO 3 Medium

6. Your portfolio has had a 15% rate of return with a standard deviation of 18% and a beta of 1.1. The average return for the S&P 500 has been 11%, and the average return for US Treasury bills has been 2%. Calculate the Sharpe ratio, Treynor ratio, and Jensen's alpha for your portfolio.

Solution:

The Sharpe ratio for the portfolio is

$$\frac{15\% - 2\%}{18\%} = 0.72$$

The Treynor ratio for the portfolio is

$$\frac{15\% - 2\%}{1.1} = 11.82$$

Jensen's alpha for the portfolio is

$$\alpha_p = 0.15 - [(0.02 + 1.1(0.11 - 0.02))] = 0.031 = 3.1\%$$

The Sharpe and Treynor ratios are reward-to-risk measures. A positive Jensen's alpha suggests that this portfolio has an abnormal return; specifically, this portfolio has a higher return than would be expected given its risk level, as measured by beta.

#### Section 15.4 LO 1 Medium

7. The monthly returns for Visa (V) and Pfizer (PFE) for 2018–2020 are provided in the chart below. In addition, the monthly return for the SPDR S&P 500 ETF Trust (SPY) is provided; SPY is often used as a proxy for the returns of the S&P 500, or a broad market index. Using Excel, calculate the arithmetic average monthly returns for V, PFE, and SPY. Also, calculate the standard deviation of returns for each of V, PFE, and SPY.

| Date   | SPY     | V       | PFE     |
|--------|---------|---------|---------|
| Jan-18 | 0.0618  | 0.0895  | 0.0226  |
| Feb-18 | -0.0364 | -0.0104 | -0.0197 |
| Mar-18 | -0.0313 | -0.0253 | -0.0135 |
| Apr-18 | 0.0092  | 0.0607  | 0.0316  |
| May-18 | 0.0243  | 0.0303  | -0.0186 |
| Jun-18 | 0.0013  | 0.0149  | 0.0196  |
| Jul-18 | 0.0417  | 0.0324  | 0.1006  |
| Aug-18 | 0.0319  | 0.0742  | 0.0398  |
| Sep-18 | 0.0014  | 0.0233  | 0.0705  |

## Principles of Finance

|        |         |         |         |
|--------|---------|---------|---------|
| Oct-18 | -0.0649 | -0.0816 | -0.0229 |
| Nov-18 | 0.0185  | 0.0280  | 0.0736  |
| Dec-18 | -0.0933 | -0.0673 | -0.0485 |
| Jan-19 | 0.0864  | 0.0233  | -0.0275 |
| Feb-19 | 0.0324  | 0.0971  | 0.0301  |
| Mar-19 | 0.0136  | 0.0563  | -0.0203 |
| Apr-19 | 0.0454  | 0.0528  | -0.0438 |
| May-19 | -0.0638 | -0.0189 | 0.0224  |
| Jun-19 | 0.0644  | 0.0774  | 0.0526  |
| Jul-19 | 0.0201  | 0.0256  | -0.1034 |
| Aug-19 | -0.0167 | 0.0158  | -0.0847 |
| Sep-19 | 0.0148  | -0.0473 | 0.0201  |
| Oct-19 | 0.0268  | 0.0398  | 0.0679  |
| Nov-19 | 0.0362  | 0.0316  | 0.0039  |
| Dec-19 | 0.0240  | 0.0201  | 0.0270  |
| Jan-20 | 0.0045  | 0.0589  | -0.0495 |
| Feb-20 | -0.0792 | -0.0865 | -0.0934 |
| Mar-20 | -0.1300 | -0.1123 | -0.0233 |
| Apr-20 | 0.1336  | 0.1092  | 0.1752  |
| May-20 | 0.0476  | 0.0924  | -0.0044 |
| Jun-20 | 0.0133  | -0.0089 | -0.1352 |
| Jul-20 | 0.0636  | -0.0143 | 0.1768  |
| Aug-20 | 0.0698  | 0.1134  | -0.0083 |
| Sep-20 | -0.0413 | -0.0553 | -0.0288 |
| Oct-20 | -0.0210 | -0.0913 | -0.0332 |

|        |        |        |         |
|--------|--------|--------|---------|
| Nov-20 | 0.1088 | 0.1576 | 0.1381  |
| Dec-20 | 0.0326 | 0.0414 | -0.0293 |

**Table 15.8** Monthly Returns for SPY, V, and PFE for 2018–2020

Solution: The results are

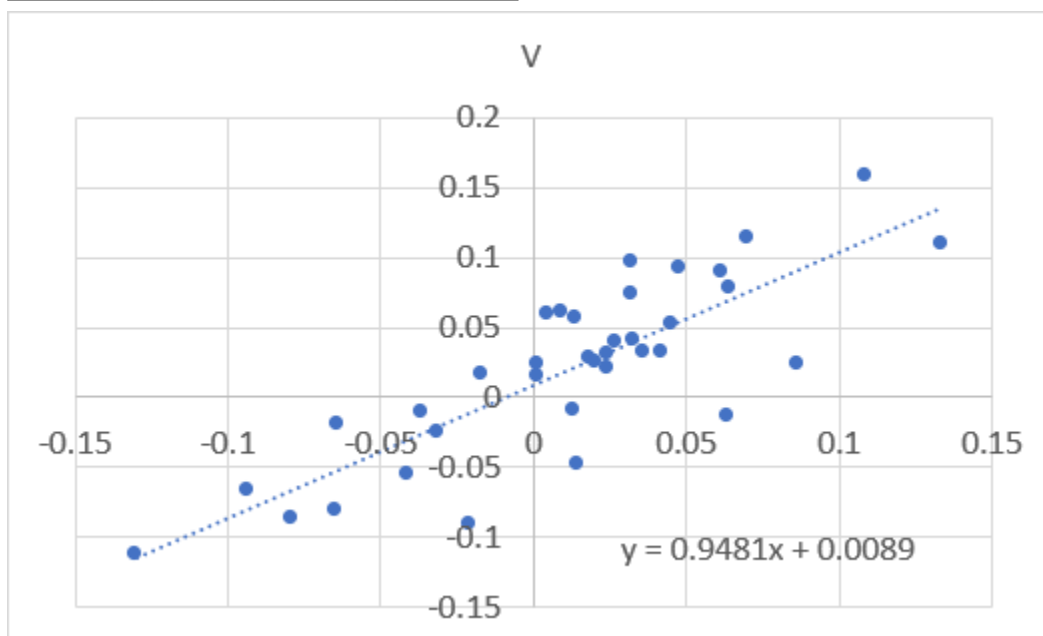
|                           | SPY    | V      | PFE    |
|---------------------------|--------|--------|--------|
| Arithmetic Average Return | 0.0125 | 0.0207 | 0.0073 |
| Standard Deviation        | 0.0553 | 0.0634 | 0.0698 |

### Section 15.5 LO 1 Difficult

8. Using the monthly returns provided in the table in problem 7, use Excel to calculate the beta for V and the beta for PFE. Which of these stocks has more systematic risk? What would you expect for the comparative returns of V and PFE?

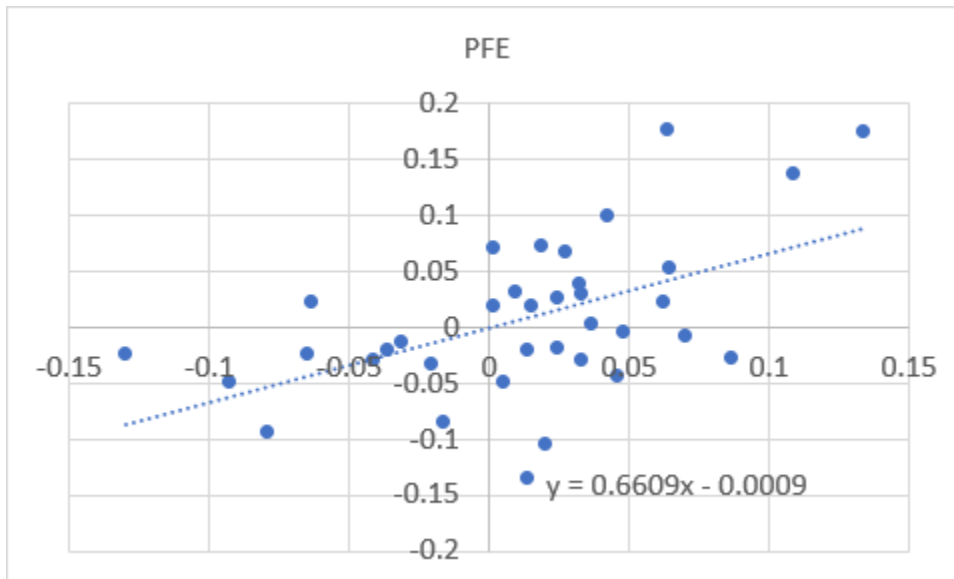
Solution: The beta for V is calculated as 0.9481.

|                           | SPY    | V      | PFE    |
|---------------------------|--------|--------|--------|
| Arithmetic Average Return | 0.0125 | 0.0207 | 0.0073 |
| Standard Deviation        | 0.0553 | 0.0634 | 0.0698 |



The beta for PFE is calculated as 0.6609.

|                           | SPY    | V      | PFE    |
|---------------------------|--------|--------|--------|
| Arithmetic Average Return | 0.0125 | 0.0207 | 0.0073 |
| Standard Deviation        | 0.0553 | 0.0634 | 0.0698 |



Because the beta for V is higher than the beta for PFE, V has more systematic risk. On average, V moves 94.81% as much as the S&P 500, while PFE moves only 66.09% as much as the S&P 500. Because both stocks have betas less than 1, the expected returns for V and PFE would both be lower than the market return. Because V has more systematic risk than PFE, investors in V should receive a higher risk premium and thus expect a higher return than investors in PFE.

### Section 15.5 LO 3 Difficult

# Chapter 15

## Test Bank

How to Think About Investing

These questions and problems can all be solved in a classroom setting using a handheld financial calculator.

### True or False Questions

1. A stock's capital gains are equal to its dividend divided by the stock price.

- A. True
- B. False

Solution: B. False

#### Section 15.1 LO 1 Easy

2. All stocks pay dividends, and therefore one can count on them having a dividend yield.

- A. True
- B. False

Solution: B. False

#### Section 15.1 LO 1 Easy

3. A risk-free asset by definition has a beta of zero.

- A. True
- B. False

Solution: A. True

#### Section 15.3 LO 1 Moderate

4. Market risk cannot be eliminated.

- A. True
- B. False

Solution: A. True

#### Section 15.3 LO 3 Moderate

## Multiple-Choice Questions

1. In measuring a stock's returns, a high standard deviation implies what?

- A. Dispersion around the stock's expected rate of return will be high.
- B. The expected rate of return on the stock is high.
- C. The level of risk in investing in the stock is low.
- D. The expected rate of return on the stock is low.

Solution: A. Standard deviation measures dispersion.

### Section 15.1 LO 2 Moderate

2. A risk-free asset will have which of the following characteristics?

- A. Investors will have no inflation risk.
- B. Investors will have a high expected return.
- C. Investors will have no default risk.
- D. Investors will have market risk.

Solution: C. Risk-free assets don't default but still have inflation risk.

### Section 15.1 LO 2 Moderate

3. Investors are said to be risk averse, which means what?

- A. They won't accept risk.
- B. They seek out risk.
- C. They have a preference for risk-free assets.
- D. They will accept risk if they are adequately compensated for it.

Solution: D. Investors are willing to take on risk if there is adequate compensation for it.

### Section 15.1 LO 3 Moderate

4. What does the benefit of diversification refer to?

- A. The combination of assets into a portfolio, creating a portfolio with a high expected return
- B. The reduction in a portfolio's standard deviation in comparison to the standard deviations of the individual assets by reducing diversifiable risk
- C. The elimination of systematic risk by combining assets into a portfolio
- D. The combination of assets into a portfolio, eliminating financial risk

Solution: B. Combining stocks in a portfolio reduces risk—but only up the point where diversifiable risk is eliminated.

### Section 15.2 LO 2 Moderate

5. Combining random stocks in a portfolio will result in what?

- A. The beta of the portfolio increasing
- B. The riskiness of the portfolio increasing
- C. The beta of the portfolio approaching 1
- D. The creation of more diversifiable risk

Solution: C. As you add stock to a portfolio, it will eventually eliminate all diversifiable risk, leaving only non-diversifiable, or market, risk.

**Section 15.2 LO 2 Moderate**

6. Why does diversification reduce overall portfolio risk?

- A. Combining stocks in a portfolio will result in the portfolio's standard deviation being brought down to the average standard deviation.
- B. The correlation coefficients of the individual stocks will cancel themselves out, leaving a portfolio with a lower standard deviation than the individual stocks' standard deviations.
- C. Diversifiable risk factors are enhanced by the combination of stocks in a portfolio, resulting in an overall reduction of risk.
- D. The combination of stocks in a portfolio will reduce the overall portfolio beta.

Solution: B. The correlation coefficients of individual stocks cancel themselves out when combined in a portfolio.

**Section 15.2 LO 2 Moderate**

7. As one adds more assets to a portfolio in order to achieve diversification, what will happen to the portfolio's risk?

- A. The risk will decline as diversifiable risk is reduced to zero, after which non-diversifiable risk will remain and the riskiness of the portfolio will stabilize.
- B. It will decline until eventually all risk is eliminated.
- C. It will increase, as the addition of riskier and riskier assets will cause overall portfolio risk to rise.
- D. It will remain the same throughout the process, as overall portfolio risk is relatively stable.

Solution: A. The combination of stock in a portfolio will reduce diversifiable risk to zero, leaving only non-diversifiable risk.

**Section 15.2 LO 2 Moderate**

8. Market risk is defined as which of the following?

- A. The level of risk that accompanies an investment in a stock
- B. An individual stock's business risk only
- C. The stock market's overall rate of return
- D. The risk that an asset will go into default

Solution: C. The market's rate of return reflects market risk.

**Section 15.2 LO 3 Moderate**

9. What is a market risk premium?

- A. The market's rate of return above that of the risk-free rate
- B. The reduction in the market's expected rate of return due to it being riskier than a risk-free asset
- C. The expected rate of return on a risk-free asset
- D. The market's expected rate of return above the rate of inflation

Solution: A. The premium referred to is the amount above the risk free rate.

**Section 15.3 LO 1 Moderate**

10. Market risk \_\_\_\_\_.

- A. cannot be eliminated, as it is non-diversifiable
- B. is synonymous with diversifiable risk
- C. is synonymous with financial risk
- D. is equal to the rate of return generated by a risk-free asset

Solution: A. Market risk is non-diversifiable.

**Section 15.3 LO 1 Moderate**

11. A stock with a beta of 1 will do what?

- A. Generate a high expected rate of return
- B. Generate a rate of return that is unpredictable
- C. Generate at least the risk-free rate of return
- D. Yield an expected rate of return equal to the market rate of return

Solution: D. The market's beta is, by definition, equal to 1.

**Section 15.3 LO 2 Moderate**

12. The expected rate of return on an asset with a beta of 1.1 will do what?

- A. Respond one for one to changes in the market's rate of return
- B. Be more sensitive to changes in the rate of return of the market
- C. Be less sensitive to changes in the rate of return of the market
- D. Generate at least the risk-free rate of return

Solution: B. A stock with a beta higher than 1 will be more responsive to changes to the market's rate of return.

**Section 15.3 LO 2 Moderate**

13. Beta calculations are subject to which of the following limitations?

- A. Beta is determined from historical data, and the stock's characteristics might have changed since the measurement was performed.
- B. Beta is known to be erratic and unpredictable.
- C. Collecting rate of return data on individual assets is difficult.
- D. The market's rate of return varies from year to year.

Solution: A. We tend to measure beta based on historical performance and then extend this history into the future.

**Section 15.3 LO 2 Moderate**

14. What does the capital asset pricing model (CAPM) calculate?

- A. The expected rate of return of an individual stock based on its overall risk
- B. The expected rate of return of an individual stock with respect to the risk-free rate of return
- C. The expected rate of return of an individual stock reflecting its financial risk
- D. The expected rate of return of an individual stock with respect to its market risk only

Solution: D. The capital asset pricing model will set an expected return for an individual asset in a portfolio setting, meaning with respect to its risk vis-a-vis the overall market.

**Section 15.3 LO 3 Moderate**

15. What does the Sharpe ratio measure?

- A. The degree to which diversifiable risk is eliminated
- B. An investor's reward in proportion to their assumption of systematic risk
- C. An investor's expected rate of return given their level of market risk
- D. How much reward an investor is getting for each unit of risk assumed

Solution: D. The Sharpe ration is a bang-for-the buck measurement.

**Section 15.4 LO 1 Moderate**

16. What does the Treynor measurement measure?

- A. The abnormal return of an asset, defined as the degree to which its actual return exceeds that predicted by the capital asset pricing model
- B. The degree to which the portfolio is optimally diversified
- C. An investor's reward in proportion to their assumption of systematic risk
- D. How much reward an investor is getting for each unit of risk assumed

Solution: C. The Treynor measurement looks at the relationship of an asset's return to systematic or market risk.

**Section 15.4 LO 2 Moderate**

17. What does Jensen's alpha measure?

- A. The degree to which diversifiable risk is eliminated
- B. The abnormal return of an asset, defined as the degree to which its actual return exceeds that predicted by the capital asset pricing model
- C. An investor's reward in proportion to their assumption of systematic risk
- D. How much reward an investor is getting for each unit of risk assumed

Solution: B. The abnormal return is defined as the excess return over and above that predicted by the capital asset pricing model.

**Section 15.4 LO 3 Moderate**

**Problems**

1. Optimus stock price started the year at \$32.00 and ended the year at \$38.00. It paid a dividend of \$1.00 per share. What is its rate of return for the year?

Solution: The capital gains yield of this stock is  $(38 - 32)/32 = 18.75\%$  and its dividend yield is  $1/38 = 2.6\%$ , so its total return is equal to  $18.75\% + 2.6\% = 21.35\%$ .

**Section 15.1 LO 1 Moderate**

2. Pessimus stock's rate of return over the past 5 years was: 5%, -2%, 10%, 12%, -8%. What was its average rate of return?

Solution: The average rate of return for this stock is

$$\frac{(.05 - .02 + .10 + .12 + .08)}{5} = 0.034 \text{ or } 3.4\%$$

**Section 15.1 LO 2 Easy**

3. What is the expected rate of return on the common stock of the Colossus Company? It has a beta of 1.2, the risk-free rate of return is 3 percent, and the market's rate of return is 10 percent.

Solution: Applying the capital asset pricing model:  $3\% + (1.2 \times (10\% - 3\%)) = 11.4\%$

**Section 15.3 LO 3 Moderate**